



DEPARTMENT OF THE AIR FORCE
HEADQUARTERS AIR FORCE SUSTAINMENT CENTER (AFMC)
TINKER AIR FORCE BASE OKLAHOMA

20 January 2026

MEMORANDUM FOR RECORD

SUBJECT: Item Description, Carl Zeiss FE-SEM replacement

1. The 76 MXSG is requesting utilization of funds for the purchase of a Carl Zeiss FE-SEM
2. The requested equipment item descriptions/specifications are outlined below.

One (1) Carl Zeiss Field Emission Scanning Electron Microscope and associated detectors for alloy analysis, fractography, and includes the latest manufacturer designs and analytical detectors.

• Specifications/Items:

- GeminiSEM 460 FE-SEM
- Carl Zeiss Software and Perpetual License for SmartSEM (No Subscriptions)
- In-lens SE detector and in-lens EsB detector
- 2x 34" Ultra widescreen LCD Monitors and Monitor holder
- Control Panel with rotary controls/keyboard (US English)
- 40nA Maximum resolution configuration
- SEM Workstation (PC) (enhanced with higher computing power: Storage, RAM, CPU, etc)
- Large PC Table (1650mm x 1000mm)
- Back Scatter Detector
- Plasma Cleaner (and associated license)
- 2x CCD/Navigation Cameras
- Support frame for 2nd PC
- Water Chiller
- Buffer pre-vacuum quiet mode
- UPS

- Active Vibration Isolation
- FE-SEM two day onsite training
- XB550L Chamber on a GeminiSEM60

• **OXFORD Analytics**

- AZtecLive Adv with Ultim Infinity 100
- Ultim Max Infinity Large Area Analytical Silicon Drift EDS Detector
- BEX detector
- EBSD (CMOS FSD with 5 diodes, Symmetry S3, and integrated AZtecHKL Adv)
- AztecCrystal MapSweeper and Strain Analysis
- HP I9 PC w/ Microsoft Office License
- 2x 27" Monitors
- Oxford installation of EDS (2), BEX, and EBSD
- Oxford advanced training (3 days)
- **Software:** SmartSEM and Zen Core, Aztec and associated modules, Microsoft Office
- **Warranty:** 2-year on-site parts and labor warranty, effective upon government acceptance.
- **Installation/Training:** Vendor(s) provide on-site installation, verification, and training (up to 10 people).
- **Quality Conformance:** Government inspection required after installation. Vendor responsible for deficiencies until government acceptance.